

Calculate the line integral of the function $\mathbf{v} = x^2\hat{\mathbf{x}} + 2yz\hat{\mathbf{y}} + y^2\hat{\mathbf{z}}$ from the origin to the point $(1, 1, 1)$ by three different routes:

(a) $(0, 0, 0) \rightarrow (1, 0, 0) \rightarrow (1, 1, 0) \rightarrow (1, 1, 1)$

(b) $(0, 0, 0) \rightarrow (0, 0, 1) \rightarrow (0, 1, 1) \rightarrow (1, 1, 1)$

(c) The direct straight line

(d) What is the line integral around the closed loop that goes out along path (a) and back along path (b)?

Solution:

The line integral has the form

$$\int_a^b \mathbf{v} \cdot d\mathbf{l}$$

where $d\mathbf{l} = dx\hat{\mathbf{x}} + dy\hat{\mathbf{y}} + dz\hat{\mathbf{z}}$

(a)

$(0, 0, 0) \rightarrow (1, 0, 0)$ has $d\mathbf{l} = dx\hat{\mathbf{x}}$ and $x: 0 \rightarrow 1, y = 0, z = 0$.

$(1, 0, 0) \rightarrow (1, 1, 0)$ has $d\mathbf{l} = dy\hat{\mathbf{y}}$ and $y: 0 \rightarrow 1, x = 1, z = 0$.

$(1, 1, 0) \rightarrow (1, 1, 1)$ has $d\mathbf{l} = dz\hat{\mathbf{z}}$ and $z: 0 \rightarrow 1, x = 1, y = 1$.

$$\int_0^1 x^2 dx + \int_0^1 2yz dy + \int_0^1 y^2 dz = \int_0^1 x^2 dx + \int_0^1 1 dz = \frac{1}{3} + 1 = \frac{4}{3}$$

(b)

$(0, 0, 0) \rightarrow (0, 0, 1)$ has $d\mathbf{l} = dz\hat{\mathbf{z}}$ and $z: 0 \rightarrow 1, x = 0, y = 0$.

$(0, 0, 1) \rightarrow (0, 1, 1)$ has $d\mathbf{l} = dy\hat{\mathbf{y}}$ and $y: 0 \rightarrow 1, x = 0, z = 1$.

$(0, 1, 1) \rightarrow (1, 1, 1)$ has $d\mathbf{l} = dx\hat{\mathbf{x}}$ and $x: 0 \rightarrow 1, y = 1, z = 1$.

$$\int_0^1 y^2 dz + \int_0^1 2yz dy + \int_0^1 x^2 dx = \int_0^1 2y dy + \int_0^1 x^2 dx = 1 + \frac{1}{3} = \frac{4}{3}$$

(c) The direct straight line is obtained by setting $x = y = z = t$ and letting $t: 0 \rightarrow 1$. Also, $d\mathbf{l} = dt\hat{\mathbf{t}}$, giving $\mathbf{v} \cdot d\mathbf{l} = 4t^2 dt$.

$$\int_0^1 4t^2 dt = \frac{4}{3}$$

(d) Since the result of the line integral of $\mathbf{v}$ is independent of the path, going from the origin to the point $(1, 1, 1)$ via the path in (a), and then from the point $(1, 1, 1)$ to the origin via the path in (b) gives

$$\underbrace{\int_0^1 x^2 \, dx + \int_0^1 1 \, dz}_{(a)} - \left[\underbrace{\int_1^0 2y \, dy + \int_1^0 x^2 \, dx}_{(b) \text{ in reverse}} \right] = \boxed{\frac{4}{3} - \frac{4}{3} = 0}$$

□
